



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	MS.Swetha Nagasri
Student Name	I.Harshika
Roll Number	2520030607
Branch	CSE
Course Outcome	CO-3

Case Study Topic: Hyderabad Metro Rail Phase-3 Expansion Planning

Work Done Summary:

Hyderabad Metro Rail Limited (HMRL) is planning a Phase-3 metro expansion to connect 6 new stations to the existing metro network through Miyapur, one of the busiest interchange stations on the Red Line. As construction cost is a critical factor, HMRL aims to design the network in a way that connects all stations with the minimum possible cost while maintaining connectivity between every station. To achieve this, the metro planning team applies Prim's Algorithm, which helps find the Minimum Spanning Tree (MST) by selecting the least-cost connections without creating cycles.

The proposed stations include Hakimpet (H), Kompally (K), Warangal Highway (W), Shamshabad (S), LB Nagar Exit (L), and Yadadri Junction (Y), along with the existing station Miyapur (M). Beginning from Miyapur, the algorithm chooses the most cost-effective connections step by step until all stations are linked. The final metro network ensures efficient connectivity with the lowest construction cost. This case study highlights how Prim's Algorithm can be used in real-world metro route planning for cost optimization and effective infrastructure development.

Implementation Details:

1. Represented the metro network as a weighted graph, where each metro station (M, H, K, W, S, L, Y) was considered a vertex and the construction costs between stations were represented as weighted edges.
2. Selected Miyapur (M) as the starting station and marked it as visited.
3. From Miyapur, the shortest available connection was to LB Nagar Exit (L) with a cost of ₹7 crore, so L was selected first.
4. From the visited stations (M, L), the shortest connection was L to Shamshabad (S) with a cost of ₹4 crore, so S was added next.
5. From the updated visited set (M, L, S), the minimum-cost connection was S to Warangal Highway (W) costing ₹6 crore, so W was selected.
6. Next, the shortest available connection was W to Kompally (K) with a cost of ₹5 crore, so K was added to the network.
7. From the remaining unvisited stations, the least-cost edge was L to Yadadri Junction (Y) with a cost of 8 crore, so Y was selected.
8. Finally, the only remaining station Hakimpet (H) was connected through Y with a cost of ₹9 crore, completing the metro network.
9. The selected connections formed the Minimum Spanning Tree (MST), and the total construction cost was calculated as ₹39 crore.

Result/Output Screenshots:

1. Prim's Algorithm connected all metro stations by selecting the minimum-cost connection at each step without forming cycles.
2. Starting from Miyapur (M), the stations were connected in the order: $L \rightarrow S \rightarrow W \rightarrow K \rightarrow Y \rightarrow H$ based on the shortest available construction cost.
3. The final network formed the Minimum Spanning Tree (MST) with a total minimum construction cost of ₹39 crore.
4. This shows that Prim's Algorithm helps design an efficient and cost-effective metro network.

File Edit Selection View Go Run Terminal Help

DS A 2 Report

EXPLORER

- DS A 2 REPORT
 - HyderabadMetroMST.java
 - MediFlowAVL.java
 - PersistentSegmentTree.java

HyderabadMetroMST.java

```

1 public class HyderabadMetroMST {
2
3     public static void main(String[] args) {
4
5         // STEP 1
6         System.out.println(x: "STEP 1\n");
7
8         System.out.println(x: "      M");

```

Run | Debug

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\iliha_eyuesh1\OneDrive\Desktop\First Year\Term 3\DSA 2\DSA 2 Report> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\iliha_eyuesh1\AppData\Roaming\Code\User\workspaceStorage\e40e96f20e973a5233789afcb9e52afb\redhat.java\jdt_ws\DSA 2 Report_9d771891\bin' 'HyderabadMetroMST'

STEP 1

```

      M
      |
      7
      |
      L

```

Total Cost = 7

STEP 2

```

      M
      |
      7
      |
      L
      |
      4
      |
      S

```

Total Cost = 7 + 4 = 11

OUTLINE

TIMELINE

JAVA PROJECTS

File Edit Selection View Go Run Terminal Help

DS A 2 Report

EXPLORER

- DS A 2 REPORT
 - HyderabadMetroMST.java
 - MediFlowAVL.java
 - PersistentSegmentTree.java

HyderabadMetroMST.java

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\iliha_eyuesh1\OneDrive\Desktop\First Year\Term 3\DSA 2\DSA 2 Report> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\iliha_eyuesh1\AppData\Roaming\Code\User\workspaceStorage\e40e96f20e973a5233789afcb9e52afb\redhat.java\jdt_ws\DSA 2 Report_9d771891\bin' 'HyderabadMetroMST'

```

      M
      |
      7
      |
      L
      |
      4
      |
      S
      |
      6
      |
      W

```

Total Cost = 7 + 4 + 6 = 17

STEP 4

```

      M
      |
      7
      |
      L
      |
      4
      |
      S
      |
      6
      |
      W
      |
      5
      |
      K

```

Total Cost = 7 + 4 + 6 + 5 = 22

OUTLINE

TIMELINE

JAVA PROJECTS

GitHub Link : <https://github.com/iliharshika/DSA-2-Project>

Student Signature	Faculty Signature